

### Description

The infant is treated with Oxygen Therapy (Extracorporeal Hyperbaric Oxygenation, Continuous ICD-10-PCS 5A05221) and IV Fluids (Introduce Electrolyte/Water Balance in Peripheral Vein, Perc ICD-10-PCS 3E0337Z.) After reviewing PCR laboratory results on February 16, 2025 at 1pm, the pediatrician updates the patient record with a diagnosis of Respiratory Syncytial Virus (RSV) Infection, documenting the updated diagnosis at 1pm.

NIST Hospital reports inpatient syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2) for all new patient admissions at least once per day. At 1:15 PM on February 16, 2015, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT^A08 Update message about this encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions.

### Comments

Dates and times are provided in this test case to illustrate the sequence of clinical and messaging events. Since the exact dates and times are not reproducible when modeling the Test Case with a Health IT Module, only date and time format will be validated within tester submitted test data.

### Pre-condition

A01-Admission message is sent before A08-Update message.

### Post-Condition

No Post-Condition

### Test Objectives

This test step examines a Health IT Module's ability to create ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements

- Setting is inpatient.
- Chief complaint is captured as narrative in PV2-3 Admit Reason
- **Patient Name is Identifiable in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in an identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Pediatric case where age is measured in months.
- Updates to Diagnoses
- Multiple Races

### Evaluation Criteria

Message Validation Report

Diagnosis code (DG1-3) validation is validated as follows:

The table below summarizes the format representations that are accepted for ONC Health IT Module Certification. Rows 1 and 2 show the formats as required by the implementation guide and standard terminology rules and best practices. The last 3 columns show acceptable alternative formats. For this test case one of the following values shall be present in the message:

#### DG1(1)-3

| Code                                                                    | Concept  | Code System | DG1-3 Message Snippet  | NIST |
|-------------------------------------------------------------------------|----------|-------------|------------------------|------|
| <b>Required Formats based on Recognized Terminology Standards Rules</b> |          |             |                        |      |
| R062                                                                    | Wheezing | I10C        | R062^<br>Wheezing^I10C | Yes  |

| 56018004                                                                                                                             | Wheezing (finding)                                   | SCT         | 56018004^Wheezing (finding)^SCT                                    | Yes  |
|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------|--------------------------------------------------------------------|------|
| <b>Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences</b> |                                                      |             |                                                                    |      |
| R06.2                                                                                                                                | Wheezing                                             | I10C        | R06.2^Wheezing^I10C                                                | Yes  |
| R062                                                                                                                                 | Wheezing                                             | I10         | R062^Wheezing^I10                                                  | No   |
| R06.2                                                                                                                                | Wheezing                                             | I10         | R06.2^Wheezing^I10                                                 | No   |
| 785744001                                                                                                                            | Bronchitis co-occurrent with acute wheeze (disorder) | SCT         | 785744001^Bronchitis co-occurrent with acute wheeze (disorder)^SCT | Yes  |
| <b>DG1(2)-3</b>                                                                                                                      |                                                      |             |                                                                    |      |
| Code                                                                                                                                 | Concept                                              | Code System | DG1-3 Message Snippet                                              | NIST |
| <b>Required Formats based on Recognized Terminology Standards Rules</b>                                                              |                                                      |             |                                                                    |      |
| R509                                                                                                                                 | Fever, unspecified                                   | I10C        | R509^Fever, unspecified^I10C                                       | Yes  |
| 416113008                                                                                                                            | Febrile disorder (disorder)                          | SCT         | 416113008^Febrile disorder (disorder)^SCT                          | Yes  |
| <b>Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences</b> |                                                      |             |                                                                    |      |
| R50.9                                                                                                                                | Fever, unspecified                                   | I10C        | R50.9^Fever, unspecified^I10C                                      | Yes  |
| R509                                                                                                                                 | Fever, unspecified                                   | I10         | R509^Fever, unspecified^I10                                        | No   |
| R50.9                                                                                                                                | Fever, unspecified                                   | I10         | R50.9^Fever, unspecified^I10                                       | No   |
| <b>DG1(3)-3</b>                                                                                                                      |                                                      |             |                                                                    |      |
| Code                                                                                                                                 | Concept                                              | Code System | DG1-3 Message Snippet                                              | NIST |
| <b>Required Formats based on Recognized Terminology Standards Rules</b>                                                              |                                                      |             |                                                                    |      |
| R0681                                                                                                                                | Apnea, not elsewhere classified                      | I10C        | R0681^Apnea, not elsewhere classified^I10C                         | Yes  |
| 724229002                                                                                                                            | Apnea of infancy (disorder)                          | SCT         | 724229002^Apnea of infancy (disorder)^SCT                          | Yes  |
| <b>Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences</b> |                                                      |             |                                                                    |      |
| R06.81                                                                                                                               | Apnea, not elsewhere classified                      | I10C        | R06.81^Apnea, not elsewhere classified^I10C                        | Yes  |
| R0681                                                                                                                                | Apnea, not elsewhere classified                      | I10         | R0681^Apnea, not elsewhere classified^I10                          | No   |
| R06.81                                                                                                                               | Apnea, not elsewhere classified                      | I10         | R06.81^Apnea, not elsewhere classified^I10                         | No   |
| 1023001                                                                                                                              | Apnea (finding)                                      | SCT         | 1023001^Apnea (finding)^SCT                                        | Yes  |
| <b>DG1(4)-3</b>                                                                                                                      |                                                      |             |                                                                    |      |
| Code                                                                                                                                 | Concept                                              | Code System | DG1-3 Message Snippet                                              | NIST |
| <b>Required Formats based on Recognized Terminology Standards Rules</b>                                                              |                                                      |             |                                                                    |      |
| R630                                                                                                                                 | Anorexia                                             | I10C        | R630^Anorexia^I10C                                                 | Yes  |
| 700213005                                                                                                                            | Avoidant restrictive food intake disorder (disorder) | SCT         | 700213005^Avoidant restrictive food intake disorder (disorder)^SCT | Yes  |
| <b>Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences</b> |                                                      |             |                                                                    |      |
| R63.0                                                                                                                                | Anorexia                                             | I10C        | R63.0^Anorexia^I10C                                                | Yes  |
| R630                                                                                                                                 | Anorexia                                             | I10         | R630^Anorexia^I10                                                  | No   |
| R63.0                                                                                                                                | Anorexia                                             | I10         | R63.0^Anorexia^I10                                                 | No   |

| 192016008                                                                                                                            | Non-organic infant feeding disturbance (disorder)   | SCT         | 192016008^Non-organic infant feeding disturbance (disorder)^SCT     | Yes  |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------|---------------------------------------------------------------------|------|
| <b>DG1(5)-3</b>                                                                                                                      |                                                     |             |                                                                     |      |
| Code                                                                                                                                 | Concept                                             | Code System | DG1-3 Message Snippet                                               | NIST |
| <b>Required Formats based on Recognized Terminology Standards Rules</b>                                                              |                                                     |             |                                                                     |      |
| R051                                                                                                                                 | Acute cough                                         | I10C        | R051^Acute cough^I10C                                               | Yes  |
| 417850002                                                                                                                            | Respiratory tract congestion and cough (disorder)   | SCT         | 700213005^Respiratory tract congestion and cough (disorder)^SCT     | Yes  |
| <b>Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences</b> |                                                     |             |                                                                     |      |
| R05.1                                                                                                                                | Anorexia                                            | I10C        | R05.1^Acute cough^I10C                                              | Yes  |
| R051                                                                                                                                 | Anorexia                                            | I10         | R051^Acute cough^I10                                                | No   |
| R05.1                                                                                                                                | Anorexia                                            | I10         | R05.1^Acute cough^I10                                               | No   |
| 49727002                                                                                                                             | Cough (finding)                                     | SCT         | 49727002^Cough (finding)^SCT                                        | Yes  |
| <b>DG1(6)-3</b>                                                                                                                      |                                                     |             |                                                                     |      |
| Code                                                                                                                                 | Concept                                             | Code System | DG1-3 Message Snippet                                               | NIST |
| <b>Required Formats based on Recognized Terminology Standards Rules</b>                                                              |                                                     |             |                                                                     |      |
| J205                                                                                                                                 | Acute bronchitis due to respiratory syncytial virus | I10C        | J205^Acute bronchitis due to respiratory syncytial virus^I10C       | Yes  |
| 55735004                                                                                                                             | Respiratory syncytial virus infection (disorder)    | SCT         | 55735004^Respiratory syncytial virus infection (disorder)^SCT       | Yes  |
| <b>Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences</b> |                                                     |             |                                                                     |      |
| J20.5                                                                                                                                | Anorexia                                            | I10C        | J20.5^Acute bronchitis due to respiratory syncytial virus^I10C      | Yes  |
| J205                                                                                                                                 | Anorexia                                            | I10         | J205^Acute bronchitis due to respiratory syncytial virus^I10        | No   |
| J20.5                                                                                                                                | Anorexia                                            | I10         | J20.5^Acute bronchitis due to respiratory syncytial virus^I10       | No   |
| 79479005                                                                                                                             | Acute bronchitis due to respiratory syncytial virus | SCT         | 79479005^^Acute bronchitis due to respiratory syncytial virus^)^SCT | Yes  |

The NIST column indicates what the NIST Tool Validation will accept in context-based testing (used for ONC Certification). A 'Yes' in the NIST column signifies that the tool will accept the format and not throw an error. A 'No' in the NIST column signifies that the tool will throw an error (specifically for DG1-3.3). However, based on the recommendation from the CDC SMEs and approval from ASTP/ONC the format is acceptable. The NIST test cases documents the instances where the test proctor can grant an exception when the tool issues and error.

#### Notes

Codes are provided in the test data for coding the diagnoses and procedures. The Tool is designed to accept those codes without generating an error. If the HIT developer uses a different but equivalent coded value within the same code system than the ones provided/accepted, the Tester may ignore errors generated by the Test Tool related to incorrect code/name of coding system when the coded data used in the message are determined via visual inspection to be valid data.

Messages in this test case will be used in a Manual Test Step under functional requirement testing.